

The Hilbert-space inverse generator problem has a negative answer

Literature-resolution packet for arXiv:1809.04743

[literature answer verified](#)

1. Exact question isolated in the source

Li–Pastor–Piskarev ask the following inverse generator question. Suppose that A generates a bounded C_0 -semigroup on a Banach space X and has dense range. Must A^{-1} generate a bounded C_0 -semigroup, or at least a C_0 -semigroup? They note that counterexamples were already known on general Banach spaces, but explicitly state that the question was still open when X is a Hilbert space.

2

Miao Li, Javier Pastor, Sergey Piskarev

Suppose the operator A on a Banach space X generates a bounded C_0 -semigroup and has dense range. Does A^{-1} generate a bounded C_0 -semigroup or at least generate a C_0 -semigroup?

It is shown in [9] that the answer is positive for generators of bounded analytic C_0 -semigroup by means of an explicit representation of the semigroup generated by A^{-1} . In fact this follows from the nice property that the class of sectorial operators is stable under inversion. For other classes of semigroups, as multiplication semigroups or contraction semigroups in a Hilbert space, the answer is positive again since the condition which characterize to be the generator of that sort of semigroup is easily shown to be stable under inversion. When the semigroup generated by A is exponentially stable, then A^{-1} is a bounded operator and accordingly it generates a C_0 -semigroup, but in general it is not uniformly bounded. This situation is analysed by an explicit representation of the semigroup generated by A^{-1} in [38, 39], including growth estimates. In [13] a sufficient condition on the resolvent map of A under which A^{-1} is the generator of a bounded C_0 -semigroup is provided. Several equivalences for A^{-1} generating a C_0 -semigroup are given in [10].

In general the answer to the question above is negative. H. Komatsu gives in [15] an example of a infinitesimal generator of a contraction semigroup on c_0 with dense range, for which the inverse operator is not an infinitesimal generator (see also [29, §1.3.6]). In [14] it is shown that the answer is negative in general for $X = \ell^p$, $1 < p < \infty$, $p \neq 2$. Recently a negative answer to the inverse generator problem in $X = L^p(\mathbb{R})$, $1 < p < \infty$, $p \neq 2$, by means of the generator of the shift group, is given in [11]. For other counterexamples see [9, 10, 38].

When X is a Hilbert space the question is still open, although there is a strong negative step in this direction. It is proved in [14] that there exist C_0 -semigroups $\{e^{tA}\}_{t \geq 0}$, of arbitrarily slow growth at infinity such that the densely defined operator A^{-1} is not the generator of a C_0 -semigroup.

In the general case, it has been shown in [33] that A^{-1} is the generator of a once-integrated semigroup whenever A is injective and generates a bounded C_0 -semigroup (A^{-1} is the generator of a regularized semigroup, see [10]).

Evidence. M. Li, J. Pastor, and S. Piskarev, *Inverses of generators of integrated fractional resolvent operator functions*, arXiv:1809.04743v1 (2018), p. 2. The full downloaded source is preserved as `source_paper.pdf`.

2. Resolving theorem

Lorist–Meyries–Veraar subsequently proved the following theorem. There are a separable Hilbert space H and a bounded operator $A \in \mathcal{L}(H)$ with dense range such that

$$\sup_{t \geq 0} \|e^{tA}\| < \infty, \quad e^{tA}x \longrightarrow 0 \quad (x \in H),$$

but the closed, densely defined operator

$$A^{-1} : \text{Ran}(A) \longrightarrow H$$

does not generate a C_0 -semigroup.

Every entry of A_n in Theorem 1.1 is given by an explicit formula. Moreover, one may take $C = 61$, see Section 2.

In the following subsections, we will formulate the questions listed in (1)-(3) in detail and describe the negative solutions that can be deduced from Theorem 1.1.

1.1. The inverse generator problem. Let X be a Banach space, suppose that A is the generator of a bounded C_0 -semigroup on X and assume that A has dense range. Then A is injective, so we may consider the operator $A^{-1} : \text{Ran}(A) \rightarrow X$, which is closed and densely defined. The inverse generator problem, posed by deLaubenfels [9], asks whether A^{-1} is again the generator of a bounded C_0 -semigroup. One may also ask the formally weaker question whether A^{-1} generates a C_0 -semigroup at all. Zwart [36] showed that these two questions are equivalent at a universal level, although they are not equivalent for a fixed generator A . The inverse generator problem is strongly motivated by numerical analysis (see Subsection 1.3), control theory (see [27]), and functional calculus (see [2, 14]).

The inverse generator problem has a positive solution if $-A$ is a sectorial operator of angle $\theta \in [0, \frac{\pi}{2})$, see [9]. On Hilbert spaces the answer is also positive for the generators of contractive C_0 -semigroups, which follows directly from the Lumer–Phillips theorem (see [36, Lemma 4.1]). In contrast, a counterexample to the general inverse generator problem on C_0 was already implicit in the work of Komatsu [21, pp. 343–344]. Further counterexamples on Banach spaces can be found in [10, 11, 16, 36].

On a Hilbert space H , Gomilko–Zwart–Tomilov constructed generators A of C_0 -semigroups with arbitrarily slow but unbounded growth and for which A^{-1} does not generate a C_0 -semigroup [16, Theorem 3]. Furthermore, if A generates an exponentially stable C_0 -semigroup on H , it was shown by Zwart [35, Corollary 2.3] (see also [3, Corollary 5.7]) that $\|e^{tA^{-1}}\| \lesssim \log(e+t)$. Thus, previous Hilbert space results either allowed the original semigroup to be unbounded or only controlled the possible growth logarithmically. The general inverse generator problem on a Hilbert space remained open.

Theorem 1.1 solves the inverse generator problem on a Hilbert space negatively, of which we record two instances in the following result.

Theorem 1.2.

- (i) (Strongly stable generator) *There exists a separable Hilbert space H and an $A \in \mathcal{L}(H)$ with dense range such that A generates a strongly stable C_0 -semigroup, i.e.*

$$\sup_{t \geq 0} \|e^{tA}\| < \infty \quad \text{and} \quad \lim_{t \rightarrow \infty} e^{tA}x = 0, \quad x \in H,$$

but the closed, densely defined operator $A^{-1} : \text{Ran}(A) \rightarrow H$ does not generate a C_0 -semigroup.

- (ii) (Exponentially stable generator) *Let $\alpha \in (0, 1)$. There exists a separable Hilbert space H and a closed, densely defined A with $0 \in \rho(A)$ such that A generates an exponentially stable C_0 -semigroup on H , i.e. there exist $M, \omega > 0$ such that*

$$\|e^{tA}\| \leq Me^{-\omega t}, \quad t \geq 0,$$

but the C_0 -semigroup $(e^{tA^{-1}})_{t \geq 0}$ is unbounded and satisfies

$$\|e^{tA^{-1}}\| \geq c (\log \log(t + e^e))^\alpha, \quad t > 0,$$

for some absolute constant $c > 0$.

Theorem 1.2(i) follows immediately from Theorem 1.1. Indeed, fix $\alpha \in (0, 1)$ and set

$$H = \bigoplus \mathbb{C}^{2^n}, \quad A = \bigoplus A_n.$$

Evidence. E. Lorist, M. Meyries, and M. Veraar, *A counterexample to the inverse generator problem and related questions*, arXiv:2608.06272v1 (2026), Theorem 1.2(i), p. 2. The full downloaded source is preserved as `resolving_paper.pdf`.

3. Exact implication and scope

Theorem 1.2(i) satisfies every hypothesis in the Hilbert-space version of the question and violates its weakest proposed conclusion:

Requirement in the question	Property of the counterexample
X is a Hilbert space	H is separable Hilbert
A has dense range	stated explicitly
A generates a bounded C_0 -semigroup	indeed strongly stable
A^{-1} is the natural inverse on $\text{Ran}(A)$	closed and densely defined
Weak conclusion: A^{-1} generates some C_0 -semigroup	false

Consequently the answer is **no**, both to bounded generation and to the weaker question of generation at all. The counterexample is stronger than the source formulation in two useful respects: A itself is bounded, and its semigroup is not merely uniformly bounded but strongly stable.

The resolving paper also proves a complementary statement in Theorem 1.2(ii): even when A generates an exponentially stable semigroup and A^{-1} does generate a C_0 -semigroup, the inverse semigroup can be unbounded. That second example independently rules out a universal boundedness conclusion; Theorem 1.2(i) is the exact theorem needed for the full negative answer.

4. Timing and classification

The source preprint was submitted in September 2018. The resolving preprint, arXiv:2608.06272v1, was submitted on 6 August 2026. Thus this packet records a very recent literature resolution rather than a new proof by the present run. The classification is therefore *literature already answered*, with the additional review caveat that the resolving source is currently a first arXiv version.

Conclusion

The Hilbert-space inverse generator problem highlighted in arXiv:1809.04743 is fully resolved in the negative by arXiv:2608.06272, Theorem 1.2(i).